Assignment P2

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1 QUESTION 1

1.1 Tasks

Table 1—Tasks performed during the hour

Task	Goal	Interface	Object
Sending a text message to a friend	Answer a friend's question	The iMessage app on my iPhone	The text message and message history
Finding and turning on an Instagram Live video from @kelseywells	Watch @kelseywells' most recent Instagram Live video	The Instagram app on my iPhone	The videos on @kelseywells's Instagram Live Page
Browsing the @thedishonhealthy's Instagram profile page	Finding a good pasta salad recipe	The Instagram app on my iPhone	The posts on @thedishonhealthy's profile page
Creating a grocery shopping list	To know what items to get at Trader Joe's	The Notes application on my iPhone	The items listed on the Notes page
Searching for Trader Joe's seasonal items via Google	To know what seasonal items to look for at Trader Joe's	The Google page on the Safari browser app on my iPhone	The search results that came from searching on Google "Trader Joe's seasonal items"

1.2 Level of Directness and Invisibility

1.2.1 *Task 1: Sending a Text Message to a Friend*

This interaction is very direct. I am typing out a text (the object) and clicking send, so directly manipulating the object; it is as straightforward as it can be. As far as invisibility, I focus on the task and not the interface because of good design. When you open up the Messages app, you click on the person you want to message, and then you start typing. It is very straightforward and mimics other real-world applications, like writing a letter.

1.2.2 *Task 2: Finding and Turning On an Instagram Live Video from @kelsey-wells*

I have never watched an Instagram live, but I had heard from a friend that @kelseywells had a good Instagram live video recently, so I wanted to find it. In terms of directness, it is relatively direct but not very direct; you have to go to her profile, then click on the Instagram Live tab, then choose the video (which is the most direct part of the interaction). There are a few steps to take and views to toggle, but it's not incredibly far from the object. In terms of invisibility, I focus mostly on the interface instead of the task. I actually had to Google first "how to find Instagram live" because I wasn't sure if that was a separate app or how to find it. Once I knew that it was on the original Instagram app, it was a matter of finding the icon on the profile page. If I did this task again it would be more invisible because the design makes more sense now, but as a novice it was unclear to me at the beginning.

1.2.3 *Task 3: Browsing the @thedishonhealthy's Instagram profile page*

This task is very direct because I can use my finger to scroll up and down on the profile page and immediately see the pasta salad recipe that I am looking for and then click on it. It feels very natural to use touch to scroll up and down. In terms of invisibility, this was invisible through good design; even when I used Instagram for the first time, I intuitively knew how to scroll through the photos because scrolling is such a common gesture for other applications.

1.2.4 *Task 4: Searching for Trader Joe's Seasonal Items via Google*

This task isn't very direct because when I search for the keywords "Trader Joe's seasonal items", this pulls up a list of articles. I first have to read the dates on the articles to see which ones are the most recent and therefore most likely to be accurate. Then, I choose an article, click on it, and read the results that the article gives. Sometimes the article is irrelevant and I have to go back to the search page and try again. Therefore, there is a gap between what I want to read (the object) and the search (the interaction).

In terms of invisibility, it has become invisible through constant practice. When I first started using Google to search for things, I was not as good at knowing what to look for (reputable websites, dates, etc.). Now, filtering results to find the relevant ones is second-nature through practice. There is good design here as well (popular websites at the top of the list, etc.) but the user still has to make some decisions about which results are most applicable to their query.

2 QUESTION 2

2.1 Task that is Invisible by Learning

One task that I do on a regular basis at work is navigate through my computer's filesystem to find and open the folders and files I need using the Terminal. When I first started it seemed very confusing, but now I don't even think about it when I change directories and open files.

2.2 Components of the Interface

When you first open the Terminal window, there isn't much available to you as the user. There is a place to type, but no information about what commands are available to you or what any of it means. Some common commands that I needed were:

- pwd (Present Working Directory) to see where you are currently located in the filesystem
- cd (Change Directory) to move either up or down in the filesystem, depending on what arguments you pass in
- ls (List) to see the folders and files available where you are currently located in the filesystem
- vi to open the file with VIM, depending on the arguments you pass in
- ESC :wq (Write Quit) to exit out of VIM

However, none of these commands are shown anywhere nor are there any buttons to press to try out new things. Especially as a beginner, this interface was very intimidating and I was concerned about breaking things.

2.3 Current Thought Process

It is no longer difficult for me to navigate through the filesystem in the Terminal and open files because I have done it so many times that it is second nature. When I navigate through the filesystem now, I actually see the filesystem in my mind's eye and can watch myself moving up and down the system.

In addition, I teach programming to adults, so I often both say the command and say the long versions of the names interchangeably when I'm speaking about

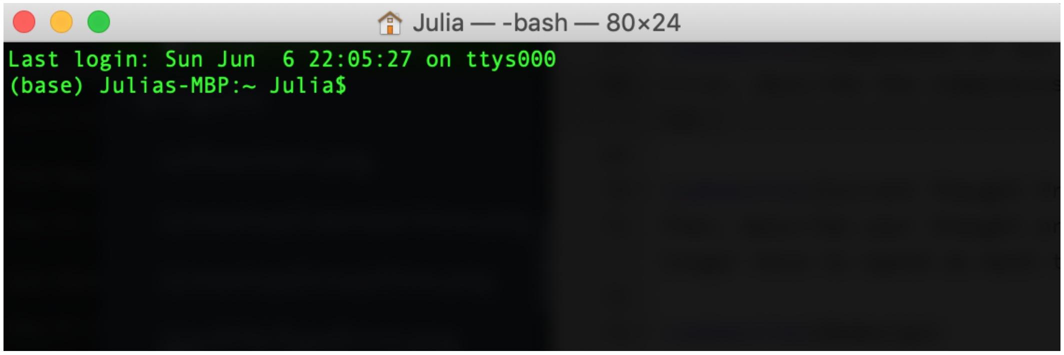

Figure 1—The Terminal window when first opened

them to my students. For example, I might say in conversation "don't forget to cd into the correct directory" or "you need to change directory first". Finally, for really tricky commands I tend to say them out loud a lot "escape colon Write Quit" so that my students can get those more difficult commands reinforced.

2.4 Redesign

One might argue that the typical "Finder" window on Macs is a better redesign of the Terminal window. However, I feel that the Terminal itself could be better designed to help new users.

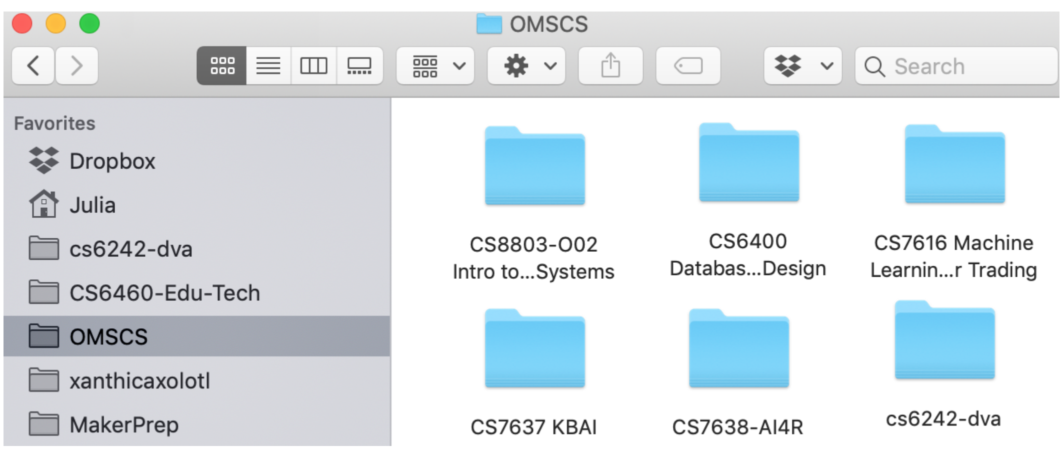

Figure 2—The Finder window on Macs

For example, it would be helpful if there was a collapsible Tree view of the current directory. There is a package called Tree that allows users to show a tree-like view, however this requires users to install it separately, and it is not collapsible. It would be great if there was a Tree view that the user could toggle from the main Terminal window.

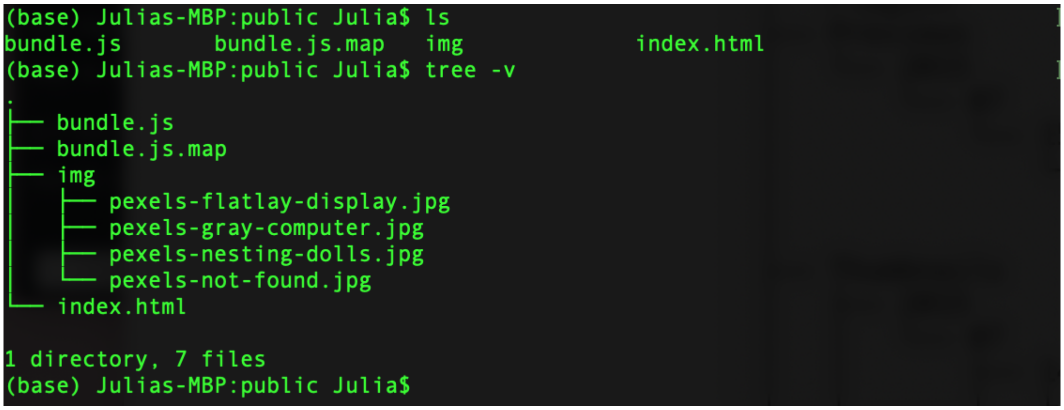

Figure 3—The Tree command to show the current directory

In addition to the Tree view, it would be great if the Tree view showed in parentheses next to each file/folder name the command for navigating to it or opening it. For example, next to the "img" branch in the Tree Terminal image, it could show instead "img (cd img)".

Finally, there could be a "commands" view on the main Terminal page that would show commonly used commands and what they do.

3 QUESTION 3

3.1 Task Domain

For this task, I will be analyzing an advanced treadmill.

3.2 Describe Feedback

3.2.1 *Visual Feedback*

There are green bars around a picture of a track on a screen that represent how far along the track the user has already run, with a flashing bar to represent where the user is currently on the simulated track.

3.2.2 *Auditory Feedback*

When the user hits mile markers, the treadmill says out loud the distance and pace, for example "1 mile in 9 minutes 42 seconds."

3.2.3 *Haptic Feedback*

As the speed increases on the treadmill, the belt moves faster which you can feel on your feet.

3.3 Design Feedback

3.3.1 *Visual Feedback*

The machine could allow you to choose a goal (for example, 3 miles in 30 minutes) and then have that goal represented on the track as a different-colored flashing bar to see if you are on pace with your goal or not.

3.3.2 *Auditory Feedback*

If your current pace is too fast or too slow for the goal you have selected, the machine could give auditory feedback on how to fix it, like "slow down to a 4.6 pace to hit your goal".

3.3.3 *Haptic Feedback*

You could be able to connect a heart rate monitor band (that goes around your chest) with the treadmill, then the treadmill could send an alert to the band to vibrate if your heart rate goes too high.

3.4 Feedback - Other

Thermoception is a perception that could be used in an interesting way. If the treadmill could detect what your current temperature is, the machine could change the fan temperature accordingly to keep your body at a good temperature. Therefore, you would know if you were getting overheated if the fan temperature became colder.

4 QUESTION 4

4.1 Tip: Emphasizing Essential Content While Minimizing Clutter

An interface that violates this suggestion is my Brother CS7000i computerized sewing machine.

4.1.1 *Description of Interface and Violation of the Tip*

This machine was designed for beginners, so it offers a lot of functionality at a low cost. However, it is also an incredibly intimidating machine for a beginner because of how many buttons and options it has. On the right side of the machine, there is a list of 70 different stitch types. This violates the rule because most (arguably 65 of the 70) of these stitch types are decorative and would be used rarely, if ever, in a normal project. The essential important stitches are not being emphasized here, especially for beginners who are just learning how to sew at all, and the whole right side of the machine is completely cluttered with information.

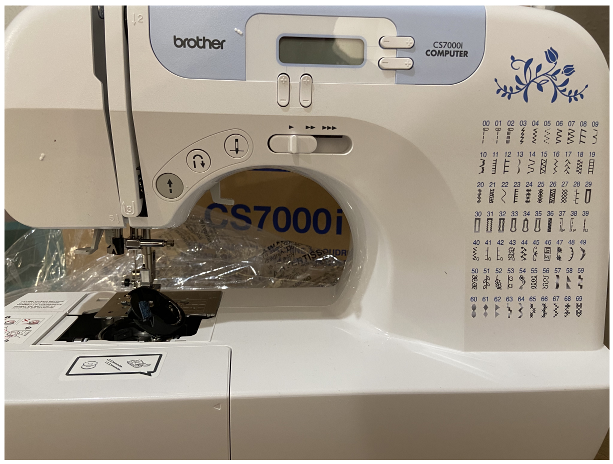

Figure 4—The Brother CS7000i sewing machine

4.1.2 *Redesign*

To redesign this with the tip in mind, it could show just the 3-5 most used stitches on the right side of the machine. This is a good reference for users who are switching stitches and don't want to look it up every time, so it would show the essential information but not the clutter. For the other approximately 65 stitches, there are a few ways the interface could deal with this. First, it could just have the other stitches in the manual for reference. This is not ideal in case the user loses the manual, but it is an option. Another option is the computerized interface at the top of the machine could include a button to scroll through the many stitch options. Finally, there is a piece of the machine at the bottom that comes out to reveal some extra tools; the other stitch types could just be listed here instead so that they are accessible but not cluttered.

4.2 Tip: Offloading Tasks from the User onto the Interface

An interface that violates this rule is the Ipsy Add-Ons page.

4.2.1 *Description of Interface and Violation of the Tip*

For users of the Ipsy beauty box subscription service, throughout the month Ipsy gives "spoilers" for what products will be available to users in the box that upcoming month. However, not all products will be available to all users during their "choice" time (when they choose what is in their box). Instead, sometimes those products are only available to those users in "Add Ons", which is when they purchase additional items after they have made their choice. The process for making the choices is often very quick, and it is easy to forget what choices you made, plus the Add Ons page is very long. Then when you get to the Add Ons screen, you can't remember what is already in your box nor can you remember which spoilers you were looking for, so you're constantly going back to previous screens to remember.

4.2.2 *Redesign*

After you have passed the choice page and made it to the Add Ons page, one way to offload this task onto the interface is to show at the top of the screen what items are already in your box so you don't look for them again. Another way to offload this task would be to star or otherwise highlight items in the Add Ons list that that were in spoilers so you can quickly scroll to find the items you wanted to purchase.